

ENVIRONMENTAL CHEMISTRY (CID CHEM 106B) LAB MANUAL



Environmental Chemistry (CID CHEM 106B)

Lab Manual

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CHAPTER OVERVIEW

1: LAB ORIENTATION

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1.1: LAB SAFETY VIDEOS

Safety Orientation Videos

Students will find below the eight Lab safety orientation videos. The first six are part of the College Lab Safety Series by the American Chemical Society. Students should watch the first seven videos before signing the safety contract and attempting the safety quiz. The eighth video is an older version of the College Lab Safety video by the American Chemical Society, which should be watched, time permitting, as it provides detailed dos and don'ts in a college lab setting.

1) ACS (American Chemical Society) College Lab Safety Series





2) Lab Safety and Techniques- Crash course safety



3) [Safety Video by American Chemical Society](#) (1991)



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1.2: LAB SAFETY CONTRACT

Student Safety Contract for Chem 3A, 1A, and 1B at Fresno City College (2)

This describes the practices required to participate and stay safe in a chemistry laboratory course at Fresno City College.

When you submit this form, it will not automatically collect your details like name and email address unless you provide it yourself.

* Required

Your Last Name *

Enter your answer

Your First Name *

Enter your answer

Your Student ID# *

The value must be a number

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1.3: Laboratory Notebooks

INTRODUCTION

Scientists are required to keep meticulous record of their work. Your notebook is a record of all the experimental information completed in class. Below are guidelines for what it should contain and what it should look like.

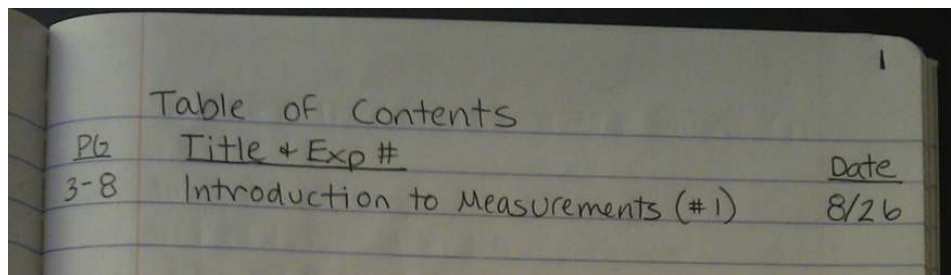
Note: Virtually all modern research and/or industry laboratory facilities practice this in one form or another. Modern pharmaceutical companies, for example, use an "Electronic Lab Notebook", which is essentially the same thing, but backed up off-site, so it is more secure and dates/times more reliable.

GENERAL INSTRUCTIONS

- Pages must be bound.
- All writing must be in un-erasable ink. (No pencils, no whiteout)
- All pages in the notebook must be numbered. (Your notebook can be pre-numbered or numbered by hand.)
- During a lab, record data and observations only in your lab notebook.
- If you make a mistake, simply strike it out with a single line. It is also a good idea to include a brief explanation of the mistake. (Ex. 2.0134 g 2.0143 g, the two last digits were accidentally switched when using the balance.)
- No pages are ever to be removed. If you really mess up a page, draw a diagonal through the page and start over on the next page (there is no penalty for doing this).

SETTING UP YOUR LAB NOTEBOOK

- Outside Front Cover: Write your name, course name, and semester.
- Inside Front Cover: Write down your name, course name, instructor's name, section, and your contact information (in case you lose your notebook).
- Pages 1-2: Leave them blank for a Table of Contents. (It should look like the example below.) As the semester goes, add the
 - Title of each experiment performed
 - The corresponding pages numbers
 - The date(s) you performed the experiment
- Pages 3-5: Reserve these pages for drawing equipment and pertinent information about the use of the equipment as instructed by your instructor.
- Pages 7-45: Record experiments.



Pg	Title + Exp #	Date
3-8	Introduction to Measurements (#1)	8/26

Figure \(\backslash\PageIndex{1}\): Example of how to fill out table of contents.

RECORDING EXPERIMENTS IN YOUR NOTEBOOK

- Start each experiment on a new page and record the following information.
- At the top of each page, write the date, title, experiment number, and page number.
- Record any important procedural notes, observations, or data recorded during your lab activities.
- If an experiment stretches over multiple days, record the date at the top of the page at the start of each day.

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CHAPTER OVERVIEW

2: The Scientific Method

Learning Objectives

- Learn to work with a group effectively in the laboratory.
- Learn how to write a testable research question or hypothesis.
- Learn to recognize different types of variables in a scientific experiment.

INTRODUCTION

In lab this week you will work with your team to conduct a very simple experiment following The Scientific Method. This activity serves as an introduction to the process of refining a vague prompt to a testable research question or hypothesis and designing and experiment to test that question or hypothesis.

In order to carry out an experiment, you will need to understand the terms Independent Variable, Dependent Variable, and Confounding (sometimes called Control) variables.

Your instructor will likely discuss this with you, here is [a short video that provides a nice explanation](#).

SAFETY PRECAUTIONS

- 1) There are no significant safety hazards in this experiment.

PROCEDURE

1. On a new page, write the date (upper right) and Title (middle) of your experiment. (Your pages should already be numbered.)
2. Write the header "Group Members" and list the names of each member of the group.
3. Write the header "Experiment Prompt" in your lab notebook.
4. As a team, select one of the following prompts for your investigation. Copy the prompt text into your lab notebook.
 - Is a tennis ball or racquetball better at bouncing?
 - How long does it take to fill a beaker with deionized water?
 - Which is bigger, a beaker or an Erlenmeyer flask?
5. Make a table in your notebook that looks like the following.

Table 1: Experiment Design Template

Dependent Variable	Independent Variable	Confounding variables
Research Question or Hypothesis		

6. Through discussions with your team, refine your prompt into a testable research question or hypothesis, including identifying relevant variables.
 - The "Dependent Variable" is what you will be measuring in your experiment.
 - The "Independent Variable" is what you will be changing about your experiment to affect the Dependent Variable.

- The "Confounding Variable" are those which could effect your result, but you don't want them to, so you will "Control" these variables to make sure they do not change.
 - Your Research Question or Hypothesis, then, can be formulated with this in mind. Make sure that it is easily tested using the above variables.
7. *STOP - before proceeding, call your instructor over to discuss and get feedback on your work so far*
 8. Write the header "Procedure" in your lab notebook.
 9. Under procedure heading, write a procedure for an experiment that will allow you to test your research question or hypothesis. Think of it like writing the step-by-step instructions you see in a cookie recipe.
 10. Leave a little space in case you need to edit or add to your procedure, then write the header "Observations" in your lab notebook.
 11. Look through your procedure and discuss what data you are planning to collect. Make a table to record this data under the observations heading.
 12. *STOP - before proceeding, call your instructor over to discuss and get feedback on your work so far*
 13. Follow your experimental procedure entering observations in the correct section. You may want to take pictures as you work for later reference or use in a presentation.
 14. Write the header "Conclusion/Discussion" then write a 2-4 sentence summary of your findings.
 15. Write the header "Member Contributions", write down the contributions made by each group member.
 16. Fill out the Table of Contents with this experiment's Page Numbers, Title, and Date.
 17. Scan and upload to the assignment on Canvas.

2.1: The Scientific Method Lab Prep Notes

The Experimental Design Template by Eric Person from his Zoo Labs curriculum has been adapted with permission under the [CC BY NC 4.0](https://creativecommons.org/licenses/by-nc/4.0/)(opens in new window) licence.

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2.1: The Scientific Method Lab Prep Notes

Supplies Needed

- 5 Tennis Balls
- 5 Racquet Balls
- Assorted beakers
- Assorted Erlenmeyer Flasks
-

Chemicals Needed

-

Other Considerations

-

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CHAPTER OVERVIEW

3: Measurements

Learning Objectives

- Learn how to make measurements in the chemistry laboratory.
- Learn how density can be used to provide evidence for the identity of an object.

INTRODUCTION

In the following experiment, you will be make length, mass, and volume measurements. You will use these to calculate the density of a piece of metal and determine it's identity.

SAFETY PRECAUTIONS

- 1) There are no significant hazards in this experiment.

PROCEDURE

Always use only pen- no pencils, no whiteout

1. On a new page, write the date (upper right) and Title (middle) of your experiment. (Your pages should already be numbered.)
2. Write the header "Group Members" and list the names of each member of the group.
3. Write the header "Length Measurement"
4. Obtain a small scrap of paper for each group member.
5. Write a short, straight pencil line in your lab notebook.
6. One at a time, using the centimeters side of your ruler, record the length of that line on your scrap of paper being careful not to let the other group member see it.
 - Measurements should always be recorded estimating one digit beyond the markings shown on the device. For a centimeter ruler, this would be "X.XX cm". Your instructor will have likely gone over this with you, but if not, [here is a video explaining the process.](#)
7. Next, on the count of 3, show each other your measurements. First, double check to see that each of your measurements are recorded to the second decimal place and they are the same, except for the last digit. (If you have any questions, call your instructor over.)
8. Record those three different measurements in your lab notebook.
9. Write the header "Density Experiment".
10. Obtain a piece of metal, called "shot" from the lab supplies.
11. Write down some observations about the piece of metal.
12. This piece of metal is one of Aluminum, Zinc, or Tin.
13. Write a hypothesis about the identity of the metal.
14. Watch [this video about using density to determine a metal's identity.](#)
15. Next, you will test your hypothesis by determining the density of your piece of metal.
 - Density is the Mass / Volume, which is unique for each substance.
16. Measure and record the mass of the metal on a laboratory balance.
 - Make sure the doors are closed and set the balance to zero (Tare).
 - Open the door and place the metal on the balance pan.
 - Record all of the digits shown on the balance and the unit.
17. Measure the volume by displacement. You will need to...
 - Obtain a 10-mL graduated cylinder

- Pour a few milliliters of water into the cylinder.
 - Observe the surface of the water closely, and write down observations in your lab notebook. Is it straight across? What shape is it?
 - Record the initial volume with unit, estimating one digit beyond the marking shown (X.XX mL). (When reading the volume of water in chemistry glassware, we always read at the bottom of the "meniscus" curve of water. Sometimes it can be difficult to see. You can hold a colored piece of paper up behind it. Here's [a very short video](#) with another technique to make it easier to see.)
 - Drop the piece of metal in, then tap the side a few times with your pen to release any bubbles.
 - Last, record the final volume, as above.
18. Now subtract the initial volume from the final volume and calculate the density.
 19. Compare your density to those in Table 1 below to find your metal's identity.
 20. This Scientific investigation was set up for you, what were your Dependent, Independent, and Confounding Variables? Fill out Table 2 in your notebook.
 21. Write a 2 - 4 sentence conclusion about your experiment.
 22. Write the header "Member Contributions", write down the contributions made by each group member.
 23. Fill out the Table of Contents with this experiment's Page Numbers, Title, and Date.
 24. Scan and upload to the assignment on Canvas.

Table 1: Density of Some Metals

Metal	Density
Aluminum	2.6 g/mL
Tin	5.8 g/mL
Zinc	7.1 g/mL
Silver	10.5 g/mL
Gold	19.3 g/mL

3.1: Measurements - Lab Preparation Notes

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3.1: Measurements - Lab Preparation Notes

Supplies Needed

- Aluminum shot
- Zinc Shot
- Tin Shot
- Ruler
- 10-mL graduated cylinder

Chemicals Needed

Other Considerations

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CHAPTER OVERVIEW

4: SEPARATION OF SAND AND SALT

Learning Objectives

The purpose of this experiment is to

- Separation of a mixture containing an unknown composition of sand and salt.
- Calculate the percentage of sand and salt in the unknown mixture.

INTRODUCTION

Matter can be classified as a pure substance or a mixture. Pure substances include elements and compounds. Elements are the simplest forms of matter that cannot be decomposed into other types of matter. They are found in the periodic table. A compound is a chemical combination of elements. Mixtures can either be homogeneous, if they are uniform throughout, or heterogeneous, if they are non-uniform.

Mixtures can be separated by physical methods such as filtration, decantation, distillation, and evaporation. In this experiment, you will use filtration and evaporation to separate a mixture of sand and salt. Once the mixture is separated, you will be asked to determine the respective percentages of sand and salt. The following example illustrates how to perform these calculations.

Sample Problem:

✓ Example $\backslash(\backslash\text{PageIndex}\{1\}\backslash)$

A mixture of sand and salt had a mass of 3.750 g. Once separated, the mass of sand was determined to be 1.873 g. The mass of recovered salt was 1.877 g. This information determines the respective percentages of sand and salt in the mixture.

Solution

To find the percentage of a component in a mixture, we use the following formula:

$$\text{Percentage of a component} = \left(\frac{\text{mass of component}}{\text{mass of mixture}} \right) \times 100$$

The percentage of sand would be

$$\text{Percentage of sand} = \left(\frac{1.873 \text{ g}}{3.750 \text{ g}} \right) \times 100 = 49.95\%$$

And the percentage of salt would be

$$\text{Percentage of salt} = \left(\frac{1.877 \text{ g}}{3.750 \text{ g}} \right) \times 100 = 50.05\%$$

SAFETY PRECAUTIONS

- 1) Wear chemical splash goggles throughout the experiment.
- 2) Gloves are provided if you wish to wear them.
- 3) Be careful when handling glassware while it is being heated and after it has been heated, as it will be hot. Allow all hot glassware to cool to room temperature before touching it.
- 4) At the end of the experiment, dispose of the sand in the "recovered sand jar" on the lab cart.

- 5) At the end of the experiment, wash the watch glass and evaporating dish containing the recovered salt in the sink with plenty of soap and water.
- 6) Be sure all glassware is clean and all equipment is returned to its proper place.

PROCEDURE

1) Using a milligram balance, weigh the following:

- an empty 50.0 mL beaker
- An empty evaporating dish with a watch glass.

2) Using a spatula, add approximately 1.00 - 2.000 g of an unknown mixture (record the mixture number or letter) to the 50 mL beaker.

3) Obtain the mass of the mixture and the beaker, and record the total mass. (Subtracting the mass of the empty beaker from the total mass will give you the mass of the mixture.)

4) Using a wash bottle, add enough deionized water to the 50 mL beaker to completely cover the mixture. Continuously stir the mixture with a stirring rod for approximately 5 minutes to dissolve the salt.

5) Decant (pour) the saltwater into the evaporating dish, keeping the sand in the 50 mL beaker.

To recover the sand, follow steps 6 through 8. To retrieve the salt, follow steps 9 through 11. Depending on the number of members in your lab group, these steps may be performed simultaneously.

6) To dry the wet sand, place the 50.0 mL beaker on a hot plate on a low setting.

Note: Once the beaker gets warm, the sand may pop. Hold the beaker with tongs and use a stirring rod to move the sand particles around it to prevent this.

7) Once the sand is completely dry and does not clump together (it will appear like sand on a beach on a hot summer day), turn off the hot plate and allow the beaker to cool. You can remove the beaker from the hot plate using beaker tongs.

8) Once the beaker is cool, obtain the mass of the 50.0 mL beaker and sand. (Subtracting the mass of the empty beaker from the combined mass of the beaker and sand will give you the mass of sand.)

9) Under the hood, set up a ring stand containing a ring clamp and wire gauze pad. Place a Bunsen burner a few inches below the wire gauze pad. (Note: Your instructor will demonstrate how to set up this apparatus. Once you set up your apparatus, be sure your instructor approves it before moving on to the next step.)

10) Place the evaporating dish on the wire gauze pad and cover it with the watch glass. Light the Bunsen burner under the wire gauze pad and heat the mixture until all the water evaporates and there is no condensation on the watch glass. Then, turn off the burner and allow the evaporating dish to cool.

11) Once the evaporating dish is cool, obtain the mass of the evaporating dish, the watch glass, and the salt. (Subtracting the mass of the empty evaporating dish and watching glass from the mass of the evaporating dish, watching glass, and salt will give you the mass of salt.)

12) Clean up all equipment, as specified in the safety precautions section.

PRE-LAB QUESTIONS

Name: _____

- 1) Briefly describe a procedure that can separate a mixture of sand and salt in three steps.
- 2) Define the following terms and provide an example of each.

Homogeneous mixture:

Heterogeneous mixture:

Element:

Compound:

- 3) A mixture weighing 5.000 g was shown to contain 2.445 g of sand and 2.555 g of salt. What is the percentage of sand and salt in the mixture?
- 4) Explain how the Law of Conservation of Matter can be applied to the previous question (Pre-lab #3).

DATA AND OBSERVATIONS

Unknown Number / Letter =

DATA TABLE

Mass empty a 50.0 mL Beaker	
Mass of the Beaker and the Mixture	
Mass of Mixture	
Mass of the Beaker and Sand	
Mass of Sand	
Mass of Empty Evaporating Dish and Watch Glass	
Mass of Evaporating Dish, Watch Glass, and Salt.	
Mass of Salt	

Percentage of Sand in the Mixture (Show Work):

Percentage of Salt in the Mixture (Show Work):

The combined Percentage of Sand and salt in the mixture (show work):

POST-LAB QUESTIONS

- 1) What are some potential sources of error in the experiment, and how can they be used to explain any discrepancies in the mixture's combined percentage of sand and salt?
- 2) Classify the following samples of matter encountered in the experiment as a homogeneous mixture, a heterogeneous mixture, an element, or a compound.

A mixture of Sand and Salt:

Sand:

Saltwater:

Salt:

Deionized water:

- 3) In this experiment, decantation separated the sand from the saltwater. What other method could have been used to achieve this separation?

[4.1: Lab Preparation Notes](#)

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4.1: Lab Preparation Notes

EQUIPMENT AND CHEMICALS NEEDED

- Milligram balance
- 50.0 mL beaker
- Stirring rod
- Hot plate
- Evaporating dish
- Watch glass
- Beaker tongs
- Spatula
- Ring stand
- Ring clamp
- Wire gauze pad
- Bunsen burner
- Deionized water in the wash bottle

* Images of equipment needed in this lab can be found in the appendix (the equipment may differ or be subject to changes; follow your instructor's directions).

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CHAPTER OVERVIEW

5: ATOMIC STRUCTURE AND FLAME TESTS

Learning Objectives

- To observe the light emitted from different metals.

PURPOSE

The purpose of this experiment is to

- Perform flame tests on solutions containing various elements, and examine the colors produced.
- Explore the emission spectrum of hydrogen.

INTRODUCTION

During the lecture, you learned about two models of atomic structure: the Rutherford and Bohr models. The Rutherford model states that the majority of the atom's mass comes from the nucleus, which is positively charged and resides at the center of the atom. In this model, electrons randomly orbit the nucleus in circular paths. The Bohr model, proposed after hydrogen's structure was examined in more detail, explains that protons and neutrons are located inside the atom's nucleus, whereas electrons exist in specific energy levels outside the nucleus.

When solutions containing various elements are heated, electrons transition from a lower to a higher energy level. Electrons eventually will return to their original energy levels, and when they do so, they often emit energy in the form of visible light. Evidence of the Bohr model can be seen by the light produced as materials containing certain elements are heated. Because electrons exist in specific energy levels and are not randomly placed outside the nucleus, each element has its characteristic pattern of energy levels and, as a result, emits its characteristic pattern of light.

As you will see in this experiment, flame tests produce a continuous, characteristic color for given elements. When viewed through a diffraction grating (a component that divides light into its constituent colors), the spectrum is discontinuous, and the colors appear as thin, vertical lines. Each vertical line corresponds to specific energy transitions for electrons in an atom of a heated element. This produces a characteristic fingerprint for the element and can be used as an identifying feature. In this lab, you will examine the emission line spectrum for hydrogen using a spectroscope containing a diffraction grating.

SAFETY PRECAUTIONS

- 1) Wear chemical splash goggles throughout the experiment.
- 2) Gloves are provided if you wish to wear them.
- 3) Make sure that no materials (book bags, purses, notebooks, etc.) are placed near open flames.
- 4) Turn off the gas immediately after working with the Bunsen burner.
- 5) Place all materials in the appropriate waste container and clean your work area when finished.

PROCEDURE

Part I: Flame Tests

- 1) Add 10 drops of solutions containing Li, Na, K, Ca, Sr, Ba, and Cu^{+2} to individual wells in a well plate.
- 2) Using a 50.0 mL beaker, obtain 10.0 mL of 1 M HCl.
- 3) Obtain a Bunsen burner with a hose and nichrome wire.

- 4) Bring all equipment (listed in 1-3) to your lab bench. Once you are ready to begin, carefully turn on/light the burner, as guided by your instructor.
- 5) Dip the nichrome wire into the HCl solution and hold the tip of the wire into the flame until it produces no color. This amounts to cleaning the wire.
- 6) Dip the tip of the wire into the lithium solution and hold it in the flame until a specific color is produced. You may have to repeat this procedure a few times to saturate the wire with the lithium solution.
- 7) Once you observe the color lithium produces in the flame, repeat steps 5-6 for the remaining elements.
- 8) When you are finished with this part of the experiment, be sure to thoroughly clean the nichrome wire and turn off the gas.

Part II. Hydrogen Emission Spectrum

Using a spectroscope, observe the hydrogen emission spectrum (which will be set up by your instructor) and record the thin, vertical lines you see in the data/observations section of this lab.

PRE-LAB QUESTIONS

Name _____

- 1) Identify the location, mass, charge, and role of the proton, neutron, and electron.
- 2) Differentiate between the Rutherford and Bohr models of atomic structure.
- 3) How can the Bohr model be applied to the colors observed from fireworks?

DATA AND OBSERVATIONS

Part I: Flame tests

Table 1: Flame Tests

Element	Color
Lithium	
Sodium	
Potassium	
Calcium	
Strontium	
Barium	
Copper	

Part II: Hydrogen emission spectrum

Replicate the observed lines in the hydrogen emission spectrum below (draw a vertical line by the indicated colors)

Red Orange Yellow Green Blue Violet

POST-LAB QUESTIONS

- 1) What was the biggest takeaway from performing this experiment?
- 2) Many streetlights are orange/yellow. Based on the results from part I, what element may be present in streetlights?
- 3) Based on the results from part I, which element(s) may be present in colors produced by the following fireworks?

Part A: Violet

Part B: Crimson red

- 4) How many vertical lines did you observe in the hydrogen emission spectrum? What do these vertical lines indicate?
- 5) How might the hydrogen emission spectrum differ from that of another element, such as neon?

[5.1: Lab Preparation Notes](#)

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5.1: Lab Preparation Notes

EQUIPMENT* AND CHEMICALS NEEDED

- Well plate
 - Nichrome wire
 - Bunsen burner with hose
 - Matches or lighter or striker
 - 50.0 mL beaker
 - Hydrogen emission spectrum tube
 - Handheld spectroscope
 - Dilute solutions (approximately 0.5 M) of either the chlorides or nitrates of Li, Na, K, Ca, Sr, Ba, and Cu^{+2}
 - 1.0 M HCl
-

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CHAPTER OVERVIEW

6: ACIDS AND BASES

PURPOSE

The purpose of this experiment is to

- Determine the pH of various solutions.
- Examine how pH affects litmus paper and red cabbage indicator.
- Identify solutions as acidic, basic, or neutral.

INTRODUCTION

pH is a measure of the hydrogen ion concentration in a solution. We can use pH values to identify whether a solution is acidic, basic, or neutral. The pH scale ranges from 0-14. Acidic solutions have a pH of less than 7, neutral solutions have a pH of 7, and basic solutions have a pH greater than 7.

In this experiment, we will first use litmus paper to determine whether a solution is acidic or basic. We will then use pH paper and a pH meter to measure the pH of the solution. Finally, we will explore the effect of pH on the red cabbage indicator, which changes colors at various pH levels.



SAFETY PRECAUTIONS

- 1) Always wear chemical splash goggles when working on this experiment.
- 2) Gloves are provided, should you wish to use them.
- 3) Dispose of all waste, as indicated by your instructor.
- 4) Thoroughly clean your work area when finished and return all equipment and chemicals to their appropriate place.

PROCEDURE

- 1) Obtain a well plate and add 10 drops of each solution listed in the equipment needed section (or refer to the data table) to individual wells.
- 2) Determine what color each solution turns the red litmus paper. To do this, dip the end of a strip of paper into the solution in the first well. To avoid wasting the litmus paper, tear off the part that was previously used and dip the remaining end into the next solution. Repeat this process for each solution. (You can likely test 3-4 solutions with a single strip of litmus paper.). In the data table, record what color the solutions turn the red litmus paper.
- 3) Repeat step 2 using blue litmus paper.
- 4) Repeat step 2 using pH paper. In this case, you could match the color of each solution turning the pH paper to the pH chart provided with the pH paper.
- 5) Determine the pH of each solution using a pH meter. Depending on the pH meter you use, specific instructions may be available. Consult your instructor for more information. Rinse the pH probe between solutions with deionized water, then gently pat dry it with a Kim wipe before immersing it in the next solution.
- 6) Add 2-3 drops of red cabbage indicator to each solution. Stir each solution with a stirring rod (rinsing and wiping the stirring rod between solutions), and record in the data table the color each solution turns.
- 7) Pour the contents of the well plate into the sink and rinse with plenty of water.
- 8) Thoroughly wash all glassware with soap and water and return all equipment and chemicals to the appropriate place. Clean your work area before leaving the lab.

PRE-LAB QUESTIONS Name _____

- 1) What color does red litmus paper turn in acidic and basic solutions?
- 2) What color does blue litmus paper turn in acidic and basic solutions?
- 3) List three properties of acids.
- 4) List three properties of bases.
- 5) What is an indicator? How can it be used to detect changes in the pH levels of solutions?

DATA AND OBSERVATIONS

Complete the following table. In the column headed “Color red litmus paper turns,” “Color blue litmus paper turns,” and “Color solution turns with red cabbage indicator,” record the color observed. Under pH paper and pH meter, record the pH value.

Table 1: Observations

	Color Change of Red Litmus Paper	Color Change of Blue Litmus Paper	pH value from pH paper	pH value from pH meter	Color Change with Red Cabbage Indicator	Is the solution acidic, basic, or neutral?
Tap water.						
Deionized water						
Vinegar						
Baking soda						
Bleach						
Detergent						
Ammonia						
Soft drink						
Glass cleaner						
Rubbing alcohol						

POST-LAB QUESTIONS

- 1) Rank the following (litmus paper, pH paper, pH meter, red cabbage indicator) from least to most accurate in determining the pH value of a solution.
- 2) What color does the red cabbage indicator turn in acidic solutions?
- 3) What color does the red cabbage indicator turn in basic solutions?
- 4) Which tests (litmus paper, pH paper, pH meter, red cabbage indicator) were used to determine whether the solution was acidic, basic, or neutral?
- 5) Did any of the test results surprise you? If yes, which ones, and why?
- 6) Provide three examples of acids and three examples of bases used in everyday life that were not explored in this experiment.

6.1: Lab Preparation Notes

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6.1: Lab Preparation Notes

EQUIPMENT AND CHEMICALS NEEDED

- Well plate
- pH meter
- Deionized water in a water bottle
- Red litmus paper
- Blue litmus paper
- pH paper
- Kimwipes
- Stirring rod
- Red cabbage indicator
- Tap water
- 0.5 M - 1 M solutions of vinegar, baking soda, bleach, detergent, ammonia
- Soft drink
- Glass cleaner
- Rubbing alcohol

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CHAPTER OVERVIEW

7: CHEMICAL REACTIONS AND EQUATIONS

PURPOSE

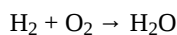
The purpose of this experiment is to:

- Perform several chemical reactions and record observations.
- Balance chemical equations.

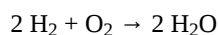
INTRODUCTION

Chemistry is defined as the study of matter and its changes. Changes in matter can be classified as physical or chemical. A physical change occurs when different forms of the same material are produced. Phase changes (such as melting or boiling) and solution formation (like the dissolution of sodium chloride in water to form saltwater) are examples of physical changes. Physical changes can be reversed without having to run a chemical reaction.

Chemical changes produce new materials. In a chemical change, atoms are rearranged as reactants (the starting materials) and are converted into products (the new materials). Chemical changes can be represented by chemical equations. Consider the conversion of hydrogen and oxygen into water:



When a chemical reaction occurs, matter cannot be created or destroyed, meaning that the same number and type of atoms must be shown on each side of the equation. To illustrate this, we must balance the equation by adding a whole number of coefficients before each element or compound. We can balance the above equation as follows:



The total number of atoms on each side of the equation can be found by multiplying the added coefficient by the given subscript. On the left side of the equation, we see four hydrogens (2×2) and two oxygens (1×2). The right side of the equation also has four hydrogens (2×2) and two oxygens (2×1). Since this equation now has the same number of atoms on each side, it is considered balanced. For more information on balancing equations, refer to the course lectures and notes.

In this experiment, you will be recording observations from ten chemical reactions. Observations will include evidence of a reaction, such as the formation of a gas, precipitate, or odor, as well as changes in color or temperature. Then, you will be asked to balance each equation associated with the given reactions.



SAFETY PRECAUTIONS

- 1) Always wear chemical splash goggles while working on this experiment.
- 2) Gloves will be provided, should you choose to wear them. Be sure to thoroughly wash your hands after completing this experiment.
- 3) Exercise caution when working with Bunsen burners. Keep notebooks, purses, books, and other belongings away from open flames. When heating test tubes, follow all directions provided by your instructor.
- 4) Thoroughly read and reread labels on bottles to make sure you are mixing appropriate solutions.
- 5) Discard chemical waste into appropriate waste containers as indicated by your instructor.
- 6) When you finish the experiment, thoroughly clean your work area and return the chemicals and equipment to their original locations.

PROCEDURE

1) Gather the following equipment:

- Bunsen burner with hose
- Matches or lighter
- Test tube holder
- Test tube rack
- 25 cm strip of copper wire
- Medium size test tube containing a small scoop of solid NaHCO_3 (baking soda)
- Medium size test tube containing a small scoop of solid $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Note: Rinse and dry your spatula between obtaining the two solids to avoid cross-contamination.

2) Carefully light the Bunsen burner, following all instructions provided by your instructor. Hold the tip of the copper wire into the flame for 10-15 seconds. Remove the wire, note any changes in the part of the copper wire placed in the flame, and record your observations.

3) Using a test tube holder, place the tip of the test tube containing NaHCO_3 into the flame for approximately 10-15 seconds, as directed by your instructor. Hold the test tube at an angle, ensuring the open end is not pointed toward you or your lab partners. Record any observations.

4) Using a test tube holder, place the tip of the test tube containing $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ into the flame for approximately 10-15 seconds, as directed by your instructor. Hold the test tube at an angle, ensuring the open end is not pointed towards you or your lab partners. Record any observations.

5) Turn off the burner. Dispose of the copper wire in the solid waste container. Pour the remaining solids in the test tubes into the inorganic waste container. Clean your work area before proceeding to the next step of the experiment.

6) Add a small scoop of solid Na_2CO_3 to a medium-sized test tube. Place the test tube in a test tube rack and add 10 drops of 1 M HCl. Record your observations. Pour the contents of the test tube into the inorganic waste container.

7) Obtain a well plate and add 5-10 drops of each solution to the following wells:

Well 1: NaCl and AgNO_3

Well 2: CaCl_2 and Na_3PO_4

Well 3: BaCl_2 and Na_2SO_4

Well 4: FeCl_3 and KSCN

Well 5: CuSO_4 and NaOH

Well 6: FeCl_3 and NaOH

Record your observations. Pour the contents of the well plate into a 600 mL beaker. Then, pour the contents of the 600 mL beaker into the inorganic waste container.

8) Thoroughly clean all glassware with soap and water. Return all equipment and chemicals to their appropriate places. Clean your work area before leaving the lab.

PRE-LAB QUESTIONS Name _____

1) Briefly define the following terms:

Melting:

Freezing:

Evaporation:

Condensation:

Sublimation:

Deposition:

2) Would the changes defined above be considered physical or chemical? Explain.

3) In a chemical reaction, the physical states of reactants and products are often given in parentheses (as you will see in this experiment). What do the following terms represent?

(s) :

(l) :

(g) :

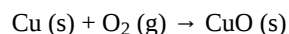
(aq) :

4) What is a precipitate? Which of the notations in pre-lab question 3 can be used to represent the formation of a precipitate?

DATA AND OBSERVATIONS

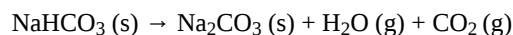
For each reaction, **balance** the provided equation and record **observations and evidence** of the reaction.

Reaction 1: Copper wire and heat



Observations:

Reaction 2: NaHCO₃ and heat



Observations:

Reaction 3: CuSO₄ · 5H₂O and heat



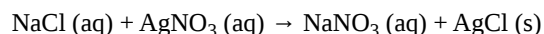
Observations:

Reaction 4: Na₂CO₃ and HCl



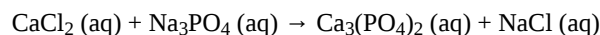
Observations:

Reaction 5: NaCl and AgNO₃



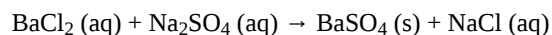
Observations:

Reaction 6: CaCl₂ and Na₃PO₄



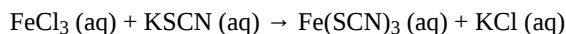
Observations:

Reaction 7: BaCl₂ and Na₂SO₄



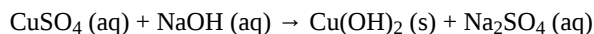
Observations:

Reaction 8: FeCl_3 and KSCN



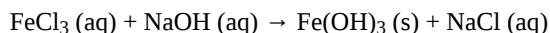
Observations:

Reaction 9: CuSO_4 and NaOH



Observations:

Reaction 10: FeCl_3 and NaOH



Observations:

POST-LAB QUESTIONS

- 1) Refer to reaction 1. Explain why oxygen is a reactant when copper wire is heated.
- 2) Refer to reaction 4. Imagine the reaction was performed in the presence of a lit match. What would you predict to happen?
- 3) Refer to reaction 8. Account for the color change observed during this reaction.
- 4) **Classify** the following changes as **physical** or **chemical**.

Part A: A match is lit.

Part B: Isopropyl alcohol is dissolved in water to form a solution of rubbing alcohol.

Part C: A piece of bread begins to mold.

Part D: When sodium metal is added to water, hydrogen gas is produced.

Part E: Steam condenses on a mirror during a hot shower.

5.) Provide three examples of physical changes and three examples of chemical changes that you encounter in daily life. Write a balanced equation to describe one of the chemical changes you describe.

7.1: Lab Preparation Notes

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7.1: Lab Preparation Notes

EQUIPMENT* AND CHEMICALS NEEDED

- Bunsen burner with hose
- Well plate
- 600 mL beaker
- Matches or lighter or Striker
- Spatula
- Wire cutters or scissors
- Test tube rack
- Test tube holder
- 3 medium-sized test tubes
- 25 cm strip of copper wire
- Solid NaHCO_3
- Solid $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- Solid Na_2CO_3
- 1 M HCl
- 0.5-1 M solutions of NaCl, AgNO_3 , CaCl_2 , Na_3PO_4 , BaCl_2 , Na_2SO_4 , FeCl_3 , KSCN, CuSO_4 and NaOH
- Solid waste container
- Inorganic waste container

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CHAPTER OVERVIEW

8: SOLUTIONS- SOLUBILITY AND CONDUCTIVITY

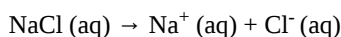
PURPOSE

The purpose of this experiment is to:

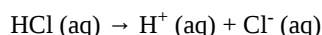
- Explore the conductivity of various solutions.
- Predict whether a substance acts as a strong electrolyte, weak electrolyte, or nonelectrolyte.

INTRODUCTION

When substances dissolve in water, they can act as strong electrolytes, weak electrolytes, or nonelectrolytes. Strong electrolytes completely ionize in water. Examples include soluble ionic compounds and strong acids (HCl, HBr, HI, HNO₃, H₂SO₄, HClO₄, HClO₃). Consider the dissolution of sodium chloride, as shown below:



When NaCl dissolves in water, it completely breaks apart into its constituent cation (Na⁺) and anion (Cl⁻). Similarly, when HCl forms a solution with water, we assume that 100% of the molecule will ionize as follows:



Weak electrolytes include weak acids (any acid other than the seven listed above) and weak bases. These substances only partially ionize in water. For example, when acetic acid, HC₂H₃O₂, dissolves in water, some of the molecules will ionize into H⁺ and C₂H₃O₂⁻, but most of what is present in aqueous solution will remain as HC₂H₃O₂.

Nonelectrolytes will not ionize in water and will remain entirely as the molecule. Examples include soluble covalent compounds. When sugar, C₁₂H₂₂O₁₁, is added to water, it will exist as C₁₂H₂₂O₁₁ and will not break apart.

The ions present in the solution allow an electrolyte to conduct electricity. The greater the number of ions present in the solution, the greater the conductivity. Strong electrolytes are better conductors than weak electrolytes since they have more ions present in the solution. Nonelectrolytes will not act as conductors because they do not ionize.

One way to determine how a substance dissolves in water is by measuring the conductivity of the solution that forms. Some conductivity meters show numerical values. The larger the value, the greater the conductivity. Others use a light to indicate conductivity. The brighter the light, the greater the conductivity. In this experiment, you will use battery-operated conductivity meters to measure the conductivity of 12 different solutions/substances. A bright green light means strong conductivity and a red light indicates no conductivity. Based on the light produced, you will then infer whether the solution acts as a strong electrolyte, weak electrolyte, or nonelectrolyte.



SAFETY PRECAUTIONS

- 1) Always wear chemical splash goggles while working with chemicals in this experiment.
- 2) Use the conductivity meters carefully and rinse the electrodes after each use in each solution and when you are finished with the experiment.
- 3) Dispose of all waste, as directed by your instructor, and thoroughly clean your work area and all equipment used.

PROCEDURE

- 1) Obtain a well plate and half-fill the wells with the following solutions: NaCl, CaCl₂, HCl, HC₂H₃O₂, NaOH, NH₃, NaHCO₃, C₁₂H₂₂O₁₁, ethanol, Gatorade, tap water, and deionized water. To do this, you could use the droppers from the bottles in which the solutions are placed or disposable plastic droppers.

- 2) Test the conductivity of each solution using the conductivity meter. Record the color of light produced in the data table. Be sure to rinse the electrodes on the conductivity meter with deionized water between each solution used.
- 3) Pour the contents of the well plate into the sink and wash with plenty of soap and water.
- 4) Thoroughly clean your work area and all equipment used. Return equipment to the appropriate place before leaving the lab.

PRE-LAB QUESTIONS Name _____

- 1) In your own words, define the following terms:

Strong Acid:

Weak Acid:

Strong Base:

Weak Base:

Ionic Compound:

Covalent Compound:

Solvent:

Solute:

Solution:

- 2) Classify the following compounds as ionic or covalent.

$\text{Ca}(\text{NO}_3)_2$:

AlCl_3 :

$\text{C}_6\text{H}_{12}\text{O}_6$:

NH_3 :

- 3) Classify the following compounds as Strong Acid, Weak Acid, Strong Base, or Weak Base.

HBr :

H_2SO_4 :

NH_3 :

KOH :

HF :

- 4) When a solution is tested with a battery-operated conductivity meter, a bright green light is shown. Based on this observation, would you predict the solution to act as a strong electrolyte, a weak electrolyte, or a nonelectrolyte?

DATA AND CALCULATIONS

Under the heading “Results of Conductivity Test,” describe the color of the light bulb. For classification of solution, write strong electrolyte, weak electrolyte, or nonelectrolyte.

Table 1: Observations

Solution	Results of the Conductivity Test	Classification of Solution
NaCl		
CaCl ₂		
HCl		
HC ₂ H ₃ O ₂		
NaOH		
NH ₃		
NaHCO ₃		
C ₁₂ H ₂₂ O ₁₁		
Ethanol		
Gatorade		
Tap water		
Deionized water		

POST-LAB QUESTIONS

- 1) Summarize the results of the experiment concerning the color of the lightbulb turned in the presence of different types of solutions.
- 2) Account for differences in the results of the conductivity test on tap water and deionized water.
- 3) Vinegar is made up of 5% acetic acid (HC₂H₃O₂) in water. Based on the results of this experiment, what would you expect to see if vinegar is tested with the conductivity meter?
- 4) Explain the results of the conductivity test on Gatorade.
- 5) Which solution would you expect to have a higher conductivity: NaCl or CaCl₂? Explain.

8.1: Lab Preparation Notes

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8.1: Lab Preparation Notes

EQUIPMENT* AND CHEMICALS NEEDED

- Well plate
 - Disposable plastic droppers
 - Deionized water
 - Ethanol
 - Gatorade
 - Tap water
 - 0.5 M - 1 M solutions of NaCl, CaCl₂, HCl, HC₂H₃O₂, NaOH, NH₃, NaHCO₃, C₁₂H₂₂O₁₁ (Be sure the concentration of each solution is the same.)
 - Battery-operated conductivity meter with light bulb
-

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CHAPTER OVERVIEW

9: INTERMOLECULAR FORCES

PURPOSE

The purpose of this experiment is to:

- Identify intermolecular forces present in the given liquids.
- Test the solubility of various liquids in water.
- Explore the evaporation of liquids and relate the results to the strengths of intermolecular forces.

INTRODUCTION

Intermolecular forces are attractive forces between atoms and molecules in a substance. The three most important intermolecular forces are London dispersion forces, dipole-dipole, and hydrogen bonding. We first look at the polarity to determine the primary intermolecular force present in a substance. Polar substances will have dipole-dipole or hydrogen bonding as the primary intermolecular force, and non-polar substances will have London dispersion forces.

Intermolecular forces play a pivotal role in determining the physical properties exhibited by substances. For instance, they can explain why certain substances form solutions with others. Understanding these forces also enables us to predict trends in boiling points and evaporation rates of liquids based on their strengths.

In this experiment, you will first identify the most important intermolecular force in various liquids by examining their structure and polarity. You will then be asked to perform a literature search to determine the boiling point of each liquid. Finally, you will explore how intermolecular forces influence solution formation and evaporation.

SAFETY PRECAUTIONS

- 1) Always wear chemical splash goggles while working on this experiment.
- 2) Gloves are provided for you if you wish to wear them.
- 3) Work with all chemicals in a well-ventilated area or under a fume hood.
- 4) At the end of the experiment, dispose of all used chemicals in the appropriate waste container.
- 5) Wash and return all glassware to the appropriate place.
- 6) Thoroughly clean your work area before leaving the lab.
- 7) Wash your hands as soon as you leave the lab.

PROCEDURE

Part A: Properties of liquids

For each of the liquids used in this experiment, draw the appropriate Lewis structure, predict whether the substance is polar or non-polar, determine the molar mass, identify the most important intermolecular force present, and do a literature search to find the boiling point (in °C). Record all this information in the data table.

Part B: Solubility

Obtain seven small test tubes and a test tube rack. To each test tube, add 1 mL of deionized water. Then, add 1 mL of the given liquids to the following test tubes:

Test tube 1 = methanol

Test tube 2 = ethanol

Test tube 3 = isopropyl alcohol

Test tube 4 = 1-butanol

Test tube 5 = ethyl acetate

Test tube 6 = butyl acetate

Test tube 7 = hexane

Gently shake each test tube and determine whether each liquid is soluble in water. Record your observations in the data table. Dispose of the contents of each test tube in the non-halogenated organic waste container.

Part C: Evaporation

- 1) Half fill wells in a Well plate with each liquid (deionized water, methanol, ethanol, isopropyl alcohol, 1-butanol, ethyl acetate, butyl acetate, and hexane).
- 2) Determine the initial temperature of the deionized water using a Celsius thermometer or the temperature probe with LabQuest. Be sure to *hold the thermometer in the liquid* until the reading stabilizes.
- 3) Once the initial temperature has been recorded, remove the thermometer from the water and *hold it in the air*. Once the temperature stabilizes, *record the final temperature*.
- 4) Repeat this procedure for each liquid.
- 5) Once you have completed this procedure, pour the contents of the well plate into a 600 mL beaker and then transfer the beaker's contents into the non-halogenated organic waste container.

PRE-LAB QUESTIONS

Name _____

1. Define the following terms:

Part A: Intermolecular forces

Part B: London dispersion forces

Part C: Dipole-dipole

Part D: Hydrogen bonding

2. Match and write the correct intermolecular force (London dispersion force, dipole-dipole, or hydrogen bonding) next to each description.

Part A: Intermolecular force that is present when hydrogen is directly bonded to oxygen, nitrogen, or fluorine:

Part B: The strongest of the intermolecular forces:

Part C: The weakest of the intermolecular forces:

Part D: The most important intermolecular force present in HCl:

Part E: Intermolecular force based on a temporary attractive force:

3. What are three factors that affect the strength of intermolecular forces?

4. **Circle** the best choice (indicated in the parenthesis) for each question:

Part A: Substances with stronger intermolecular forces tend to have (**higher/lower**) boiling points and evaporate (**more readily/less readily**) than substances with weaker intermolecular forces.

Part B: Substances with similar intermolecular forces tend to be (**miscible/immiscible**) with each other.

DATA AND OBSERVATIONS

Name _____ Lab Partner(s) _____

Part A: Properties of Liquids

Table \(\PageIndex{2}\): PROPERTIES OF LIQUIDS

*Substance	Molar Mass	Boiling Point	Lewis Structure	Polar or Nonpolar	Prim Forc
Water [H ₂ O] HOH					
Methanol [CH ₄ O] CH ₃ OH					
Ethanol [C ₂ H ₅ OH] CH ₃ CH ₂ OH					
Isopropyl Alcohol [C ₃ H ₈ O] CH ₃ CHOHCH ₃					
1-butanol [C ₄ H ₉ -OH] (CH ₃ (CH ₂) ₂ CH ₂ -OH)					
Ethyl Acetate [C ₄ H ₈ O ₂] CH ₃ COOCH ₂ CH ₃					
Butyl acetate [C ₆ H ₁₂ O ₂] CH ₃ COO(CH ₂) ₃ CH ₃					
Hexane [C ₆ H ₁₄] CH ₃ (CH ₂) ₄ CH ₃					

*Name, [Chemical Formula], and Condensed Formula (Showing Connectivity).

Part B: Solubility

Table \(\PageIndex{3}\): SOLUBILITY IN WATER

Liquid	Soluble or Insoluble in Water?
1-butanol	
Butyl acetate	
Ethanol	
Ethyl acetate	
Hexane	
Isopropyl alcohol	
Methanol	

Part C: Evaporation

Table \(\PageIndex{4}\): CHANGE IN TEMPERATURE DURING EVAPORATION

Liquid	Initial Temperature	Final Thermometer Reading	Change in Temperature
Water			
Methanol			
Ethanol			
Isopropyl alcohol			
1-butanol			
Ethyl acetate			
Butyl acetate			
Hexane			

POST-LAB QUESTIONS

1. Refer to the results from part A. What is the relationship (if any) between the molar masses and boiling points of the liquids?
2. Refer to the results from part A. Do you notice a trend between the boiling points and primary intermolecular forces in the liquids?
3. Refer to the results from part B. What type of substances were soluble in water? How can you use the concept of intermolecular forces to explain this?
4. Refer to the results from part B. Explain why this was the case for each substance insoluble in water.
5. Write a summary of the results from part C.
6. Is there a connection between the observed evaporation results of the liquids in part C and the boiling points recorded in part A? Explain.
7. Based on the results from part C, is evaporation an endothermic or exothermic process? Explain.
8. Based on the results of this experiment, what can you conclude about the relationship between the strengths of intermolecular forces and the boiling points/evaporation of a liquid?

Please click here to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

[9.1: Lab Preparation Notes](#)

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9.1: Lab Preparation Notes

EQUIPMENT AND CHEMICALS NEEDED

Table 1: Equipment and Chemicals

EQUIPMENT	CHEMICALS	CHEMICALS
Well plate	Deionized water	Ethyl acetate
600 mL beaker	Methanol	Butyl acetate
7 small test tubes	Ethanol	Hexane
Test tube rack	Isopropyl alcohol	Non-halogenated organic waste container
Thermometer	1-butanol	-

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CHAPTER OVERVIEW

10: PHYSICAL AND CHEMICAL PROPERTIES

PURPOSE

The purpose of this experiment is to:

- Record the appearance of several metals and nonmetals.
- State whether a solid is soluble or insoluble in water.
- State whether a liquid is soluble or insoluble in water.
- Explain whether a substance is undergoing a physical or chemical change.
- Measure boiling points and record test tube reactions.

INTRODUCTION

Chemistry is the study of matter and any changes it undergoes. Matter is defined as anything that occupies space and has mass. Matter can be classified into three states: solid, liquid, and gas. These properties indicate that matter has specific physical characteristics, which correspond to the three **phases** of matter. A solid has a definite shape and volume. Solids are tightly packed, with atoms having a fixed volume and a definite shape. Liquids have a fixed volume but not a definite shape and take the shape of their containers. Gases have neither a definite shape nor a definite volume, and they expand to fill the space within their containers.

Matter can also be classified into mixtures and pure substances. A pure substance has a constant composition. All have the same composition and properties. Pure substances are divided into two classes: elements and compounds. Pure substances that cannot be broken down into simpler substances by chemical changes are called elements. Some examples are iron, silver, gold, aluminum, sulfur, oxygen, and copper. Pure substances that can be broken down to produce either elements or other compounds, or both, by chemical changes are called compounds.

Although there are 118 elements, tens of millions of chemical compounds result from different combinations of these elements. Each compound has a specific composition and definite chemical and physical properties that distinguish it from all other compounds. And, of course, there are innumerable ways to combine elements and compounds to form different mixtures.

Scientists use the periodic table to reference information about an element, such as atomic mass and chemical symbols. The periodic table's arrangement also allows scientists to discern trends in element properties, including electronegativity, ionization energy, and atomic radius. The periodic law states that the physical and chemical properties of the elements recur systematically and predictably when the elements are arranged in order of increasing atomic number.

The **periodic table** organizes elements based on their increasing atomic numbers, which allows for the prediction of their chemical and physical properties. Elements within the same group or column often share similar properties due to their similar outer electron configurations.

The properties that chemists use to describe matter fall into two general categories.

Physical properties are characteristics that describe matter. They include size, hardness, state, shape, color, density, melting, boiling, freezing, solubility, and mass.

Chemical properties describe how matter changes its chemical structure or composition. Flammability, toxicity, acidity, the reactivity of various types, and the heat of combustion

SAFETY PRECAUTIONS

1. Always wear chemical splash goggles during the experiment.
2. Gloves are provided for those who wish to wear them.
3. Exercise extreme caution when using a hot plate or Bunsen burner to heat water.

4. Water used for boiling point determination can be discarded in the sink. Other chemical waste solutions should be placed in appropriate waste containers as advised by the instructor.
5. Ensure all glassware is clean, and all equipment is returned to its designated place.
6. Thoroughly clean your work area and return chemicals and equipment to their designated places after the experiment.
7. Clean the lab benches and verify all equipment is in your lab drawer before leaving.
8. Wash your hands as soon as you leave the lab.

PROCEDURE

A. Identification, appearance, and observations of several metals, metalloids, and nonmetals.

Using the [virtual periodic table](#) and answering the questions in the *data table below in data and observations part A*.

1. Classify the following: Carbon, Calcium, Sodium, Nitrogen, Oxygen, Phosphorus, Neon, Vanadium, Lithium, as metals, metalloids, and nonmetals in the data sheet.
2. Identify the elements and group numbers of named groups: Alkali Metals, Alkaline earth metals, halogens, and noble gases.
3. Which group of elements is considered transition metals?
4. Which elements are considered Metalloids?

B. Physical and Chemical Properties and Changes

1. Take a 250.0 mL beaker and add approximately 175 mL of distilled water to it. Bring the mixture to a boil on a hot plate*.
2. While waiting for the water to boil, take a test tube large enough not to be immersed in the above 250.0 mL beaker and add 25 drops of alcohol, along with a boiling chip. Note the alcohol's identity. Suspend the thermometer in the test tube using a split cork. Ensure the tip of the thermometer is at least 1.0 cm above the alcohol level in the test tube. Place the test tube in the test tube rack.
3. When the water starts boiling, record the boiling point in part B of the data table and observations. Identify whether this is a chemical or physical change. After recording the boiling point, turn off the hot plate.
4. Place the test tube with alcohol into the above beaker with hot water. Record the temperature as the alcohol starts to boil. If your water has cooled down and the alcohol does not boil, turn on your hot plate to warm the water.
5. After the remaining alcohol has cooled, pour it into the organic waste container, and the water can be discarded down the drain. Clean your work area and all glassware before moving on to the next step of the experiment.
6. Obtain six test tubes and place them on a test tube rack. Add 10 mL of distilled water to each. Add half a spatula or 25 drops of the mentioned chemicals to the test tubes. Into the Test tube: 1- Sodium chloride NaCl , 2- Sodium carbonate Na_2CO_3 , 3- Calcium carbonate CaCO_3 , 4- Starch, 5- twenty five drops of ethanol (or methanol), 6- 25 drops of hexane, respectively (add only one compound into each test tube of distilled water). Swirl the liquid in the test tubes, and if necessary, use the glass rod to stir the solid in all three test tubes. Record the observations and solubilities and identify the type of change involved. Chemical waste solutions should be placed in appropriate waste containers.
7. Clean the six test tubes used above and place them on a test tube rack. Add 10 mL of 1 M HCl to each. Add half a spatula or 25 drops of the mentioned chemicals to the test tubes. Into the Test tube: 1- Sodium chloride NaCl , 2- Sodium carbonate Na_2CO_3 , 3- Calcium carbonate CaCO_3 , 4- Starch, 5- twenty five drops of ethanol (or methanol), 6- 25 drops of hexane, respectively (add only one compound into each test tube). Stir using the glass rod. (Remember to clean the glass rod after mixing it into each test tube.) Record the observations and solubilities and identify the type of change involved.
8. As instructed by your instructor, discard the contents of the test tubes into the designated waste container. Thoroughly clean all glassware with soap and water. Return all equipment and chemicals to their appropriate place. Before leaving the lab, clean the bench and your work area.

PRE-LAB QUESTIONS

Name: _____

1. This add fixed lines that are not deletable by LibreTexts bots. Can select from 1 to 10 lines to fix

2. The columns and rows of the periodic table are called:
3. The elements in the two rows at the bottom of the periodic table are called:
4. This add fixed lines that are not deletable by LibreTexts bots. Can select from 1 to 10 lines to fix

5. Define what a Chemical property is and give five examples.

DATA AND OBSERVATIONS

Name: _____ Lab Partner(s): _____

A. Identification, appearance, and observations of several metals, metalloids, and nonmetals.

1. Classify the following as metals, metalloids, and nonmetals.

Table \(\PageIndex{2}\): Classify the following as metals, metalloids, and nonmetals.

Element Name	Symbol of Element	Identify whether Metal / Nonmetal / Metalloid	Group #	State (s, l, g)
Carbon				
Calcium				
Sodium				
Nitrogen				
Oxygen				
Phosphorous				
Neon				
Vanadium				
Lithium				
Germanium				

2. Identify the elements and group numbers of named groups: Alkali Metals, Alkaline earth metals, halogens, and noble gases.

a) Alkali Metals Group Number:

Names and Symbols of Alkali Metals:

b) Alkaline Earth Metals Group Number:

Names and Symbols Alkaline Earth Metals:

c) Halogens Group Number:

Names and Symbols of Halogens:

d) Nobel Gases Group Number:

Names and Symbols of Nobel Gases:

e) Transition metals, Group Numbers:

Names and Symbols of five Transition metals:

f) Names and Symbols of five of the Metalloids:

B. Physical and Chemical Properties and Changes

1. Boiling Point:

i) Boiling Point of Water:

Identify whether the boiling point of water is a Physical or Chemical change:

ii) Boiling Point Alcohol:

Identify whether the boiling point of Alcohol is a Physical or Chemical change:

2. Determine whether a change is chemical or physical using Water.

Table \(\PageIndex{3}\): Classify Chemical or Physical Change using Water

Test tubes #	Observation	Soluble / Insoluble	Physical /Chemical Change
1. Sodium chloride + water (NaCl + H ₂ O)			
2. Sodium Carbonate + water (Na ₂ CO ₃ + H ₂ O)			
3. Calcium Carbonate + water (CaCO ₃ + H ₂ O)			
4. Starch + water			
5. 25 drops of ethanol + water or (25 drops of methanol + water)			
6. 25 drops of hexane + water			

3. Identify if a Chemical or Physical change using Hydrochloric acid.

Table \(\PageIndex{4}\): Classify Chemical or Physical change using Hydrochloric acid

Test Tubes #	Observation	Soluble / Insoluble	Physical /Chemical Change
1. Sodium chloride + HCl (aq) (NaCl + HCl (aq))			
2. Sodium Carbonate + HCl (aq) Na ₂ CO ₃ + HCl (aq)			
3. Calcium Carbonate + HCl (aq) CaCO ₃ + HCl (aq)			
4. Starch+ HCl (aq)			
5. 25 drops of ethanol + HCl (aq) (or methanol + HCl (aq))			
6. 25 drops of hexane + HCl (aq)			

POST-LAB QUESTIONS

1. Classify the following elements and complete the table below.

Table \(\PageIndex{5}\): Classify the Elements

Name of Element	Symbol of Element	Identify whether Metal / Nonmetal/Metalloid	Group Number	Period
Lithium				
	Si			
Zinc				
	Mg			
Copper				
	I			
Uranium				
	Cl			
Tin				
	Ge			

2. Identify whether a chemical or physical change.

Table \(\PageIndex{6}\): Classify the Elements

Chemical or Physical Change	
1. The density of zinc is 7.41 g/cm ³	
2. Zinc combines with Hydrochloric acid to give zinc chloride and hydrogen gas	
3. Silver tarnishes to silver oxide when exposed to moisture	
4. Melting of polar ice caps due to global warming	
5. Copper can be drawn into a thin wire and wrapped into a cord	
6. An iron nail rusts when exposed to moisture	

Please click [here](#) to access the Pre-Lab, Data Tables, and Post-Lab in [Word](#) or [PDF](#) format. Complete them and upload according to your instructor's instructions.

10.1: Lab Preparation Notes

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10.1: Lab Preparation Notes

EQUIPMENT AND CHEMICALS NEEDED

Table 1: Equipment and Chemicals

Equipment	Equipment	Chemicals
Safety goggles	Thermometer	1.0 M Hydrochloric Acid (HCl)
Spatula	Large test tube and split cork	Alcohol (ethanol/methanol)
Hot plate	Test tube holder	Sodium chloride NaCl
10 mL graduated cylinder	Medium-sized test tubes	Sodium carbonate Na_2CO_3
Glass Stirring Rod	Test tube rack	Calcium carbonate CaCO_3
250 ml beaker	Boiling chip	starch
Deionized water		Hexane

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